

A Workflow for Offline Model-Free Robotic Reinforcement Learning

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Abstract: Offline reinforcement learning (RL) enables learning control policies by utilizing only prior experience, without any online interaction. This can allow robots to acquire generalizable skills from large and diverse datasets, without any costly or unsafe online data collection. Despite recent algorithmic advances in offline RL, applying these methods to real-world problems has proven challenging. Although offline RL methods can learn from prior data, there is no clear and well-understood process for making various design choices, from model architecture to algorithm hyperparameters, without actually evaluating the learned policies online. In this paper, our aim is to develop a practical workflow for using offline RL analogous to the relatively well-understood workflows for supervised learning problems. To this end, we devise a set of metrics and conditions that can be tracked over the course of offline training, and can inform the practitioner about how the algorithm and model architecture should be adjusted to improve final performance. Our workflow is derived from a conceptual understanding of the behavior of conservative offline RL algorithms and cross-validation in supervised learning. We demonstrate the efficacy of this workflow in producing effective policies without any online tuning, both in several simulated robotic learning scenarios and for three tasks on two distinct real robots, focusing on learning manipulation skills with raw image observations with sparse binary rewards. Explanatory video and additional content can be found at sites.google.com/view/offline-rl-workflow.

Keywords: workflow, offline RL, offline tuning

1 Introduction

Offline reinforcement learning (RL) can in principle make it possible to convert existing large datasets of robotic experience into effective policies, without the need for costly or dangerous online interaction for each training run. While offline RL algorithms have improved significantly [1, 2, 3, 4, 5], applying such methods to real-world robotic control problems presents a number of major challenges. In standard online RL, any intermediate policy found during training is executed in the environment to collect more experience, which naturally allows for an evaluation of the policy performance. This ability to evaluate intermediate policies lets practitioners use “brute-force” to evaluate the effects of various design factors, such as model capacity and expressivity, the number of training steps, and so forth, and facilitates comparatively straightforward tuning. In contrast, offline RL methods do not have access to real-world on-policy rollouts for evaluating the learned policy. Thus, in order for these methods to be truly practical for real-world applications, we not only require effective algorithms, but also an effective *workflow*: a set of protocols and metrics that can be used to reliably and consistently adjust model capacity, regularization, etc in offline RL to obtain policies with good performance, without requiring real-world rollouts for tuning.

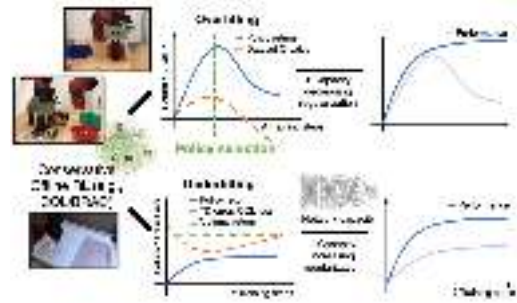


Figure 1: **Our proposed workflow** aims to detect overfitting and underfitting, and provides guidelines for addressing these issues via policy selection, regularization, and architecture design. We evaluate this workflow on **two** real-world robotic systems and simulation domains, and we find it to be effective.